SPI Ultrasound Physics — Express Review Sheets

Formulas, "if X doubles" rules, worked examples and exam traps for SPI exam review

MindShelf

How these sheets map to the published SPI content outline

Outline domain	Weight	Sections here
Perform Ultrasound Examinations	23%	Sound, attenuation, reflection, artifacts, bioeffects
Manage Ultrasound Transducers	7%	Transducers
Optimize Sonographic Images	26%	Pulsed sound, resolution, instrumentation
Apply Doppler Concepts	34%	Doppler and hemodynamics
Provide Clinical Safety & Quality Assurance	10%	Safety, quality assurance

A condensed review aid for the physics side of each domain, not a complete course: patient care, ergonomics, 3D/4D, contrast agents and infection control are not covered. Exam units: mm, µs, MHz, cm, dB, mW/cm².

Sound & acoustic parameters

Ultrasound: above 20 kHz. Longitudinal mechanical wave; needs a medium.

$$T = \frac{1}{f} \quad \lambda = \frac{c}{f} \quad c = f\lambda$$

$c_{\text{soft tissue}}$ = 1,540 m/s = 1.54 mm/µs. **Speed depends on the medium only**: stiffer (less compressible) is faster, denser is slower (same stiffness). Order: lung/air < fat < soft tissue < muscle < bone. Source sets f and T; source **and** medium set λ. Amplitude, power, intensity: set by the system; $I = P/A$, $I \propto \text{amplitude}^2$.

EXAMPLE 5 MHz in soft tissue

T = 1/5 = 0.2 µs; λ = 1.54/5 = 0.308 mm. 10 mW on 0.5 cm²: I = 20mW/cm^2.

Decibels

$$\text{dB} = 10 \log_{10}\left(\frac{I_2}{I_1}\right)$$

+3 dB: intensity doubled; −3 dB: halved; +10 dB: ×10. Example: 3 dB ⇒ ratio $10^{3/10} \approx 2$.

Pulsed ultrasound

$$\text{PD} = nT = \frac{n}{f} \quad \text{SPL} = n\lambda$$

$$\text{PRP} = \frac{2d}{c} \quad \text{PRF} = \frac{1}{\text{PRP}} \quad \text{DF} = \frac{\text{PD}}{\text{PRP}}$$

Soft tissue: PRP ≈ 13 µs/cm of imaging depth. PD and SPL: set by source (cycles, f); SPL also by medium. PRP, PRF, DF: change with **depth** (sonographer). CW: DF = 100%; imaging DF is tiny.

EXAMPLE 3-cycle pulse, 5 MHz, 10 cm depth

PD = 3/5 = 0.6 µs; SPL = 0.924 mm. PRP = 2 × 10/0.154 ≈ 130 µs; PRF max = 7.7 kHz; DF ≈ 0.46%.

Intensities

SP = spatial peak (beam center), SA = spatial average; TP = temporal peak (pulse on), TA = temporal average (includes dead time), PA = pulse average.

$$\text{BUF} = \frac{\text{SP}}{\text{SA}} \geq 1 \quad \text{DF} = \frac{\text{TA}}{\text{TP}} \leq 1$$

Highest: SPTP. Lowest: SATA. SPTA: the one tied to **tissue heating**. In CW, TA = TP.

Attenuation

Absorption (main cause, energy to heat) + reflection + scattering.

loss (dB) ≈ $\alpha \times f \times$ path α = 0.5 dB/cm/MHz

Pulse–echo path = 2 × depth. Half-value layer: 3 dB /(αf). Attenuation: lowest in water, then blood, soft tissue, muscle, bone; highest in air/lung. Rayleigh scattering (red cells): $\propto f^4$.

EXAMPLE Loss and half-value layer

5 MHz, reflector at 6 cm: 0.5 × 5 × 2 × 6 = 30 dB. At 2 MHz: HVL = 3/(0.5 × 2) = 3 cm.

Impedance, reflection & refraction

$$Z = \rho c \quad \text{(rayls)}$$

Normal incidence:

$$\text{IRC} = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2 \quad \text{ITC} = 1 - \text{IRC}$$

Reflection needs an impedance **difference**; equal Z: no echo. Refraction needs **oblique** incidence **and** different speeds:

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{c_2}{c_1}$$

Specular: smooth surface larger than λ, angle-dependent. Scattering: rough or smaller than λ.

EXAMPLE Echo and bend

Z_1 = 1.5, Z_2 = 1.7 MRayl (exercise values): IRC ≈ 0.39%. $\sin\theta_i$ = 0.5, c_1 = 1,540, c_2 = 1,600 m/s: $\sin\theta_t \approx 0.519$, bent away from the normal ($c_2 > c_1$).

Resolution

Axial (LARRD: longitudinal, axial, range, radial, depth):

$$\text{axial} = \frac{\text{SPL}}{2} = \frac{n\lambda}{2}$$

Better (smaller number): higher f, fewer cycles (backing). Fixed for a given pulse. **Lateral** (LATA) = beam width; best at the focus. Usually worse than axial. **Elevational** = slice thickness. **Temporal** = frame rate:

$$\text{FR} = \frac{c}{2dNF}$$

N lines per frame, F focal zones: deeper, more lines or more foci lower the frame rate.

EXAMPLE Axial and frame rate

3 cycles at 5 MHz: axial = SPL /2 = 0.462 mm. 10 cm, 128 lines, 1 focus: FR ≈ 60 Hz.

Transducers

Piezoelectric effect (PZT, lead zirconate titanate): voltage ↔ deformation, both ways. Heating above the Curie point depolarizes the element. Pulsed frequency: f = c_{element}/(2 thickness); thinner, faster element: higher f. CW frequency = drive voltage frequency. **Matching layer**: $\lambda/4$ thick, impedance between element and skin; gel removes air. **Backing** (damping): shorter pulse, better axial resolution, wider bandwidth, lower Q factor, lower sensitivity. Near zone length:

$$\text{NZL} = \frac{D^2}{4\lambda} = \frac{D^2 f}{4c}$$

At the end of the near zone, beam width ≈ $D/2$; at 2 NZL, ≈ D. Arrays: linear (rectangle), curvilinear (blunted sector), phased (sector; electronic steering and focusing), annular (focus in 2 planes, mechanical steering).

EXAMPLE Element and near zone

Exercise values: c = 4 mm/µs, 0.4 mm thick ⇒ f = 5 MHz. D = 6 mm at 5 MHz: NZL ≈ 29 mm.

Instrumentation & processing

Receiver order: amplification (gain), compensation (TGC), compression, demodulation, reject. **Gain** changes received echoes only; **output power** changes the transmitted beam and patient exposure. Image too dark: raise gain first (ALARA). Pre-processing (before storage, cannot be undone on a frozen image): TGC, write zoom, persistence. Post-processing: read zoom, contrast/gray map, measurements on frozen images. Dynamic range: ratio largest/smallest signal (dB); compression narrows it. Gray shades = 2^{bits}: 8 bits ⇒ 256. Harmonic imaging: receives $2f_0$; fewer side-lobe and reverberation artifacts. Spatial compounding: averaged frames from several angles; less speckle and shadowing.

Doppler

$$\Delta f = \frac{2 f_0 v \cos\theta}{c}$$

cos 0° = 1 (largest shift); cos 90° = 0 (no shift). Shift in the audible range. Pulsed wave (PW): range resolution, aliasing possible. Continuous wave (CW): no aliasing, no range resolution. Nyquist limit = PRF /2. Aliasing when Δf > PRF /2.

$$v_{\text{max}} = \frac{\text{PRF} \times c}{4 f_0 \cos\theta}$$

Fix aliasing: raise scale (PRF), move the baseline, lower f_0, shallower gate, CW. Color Doppler: mean velocity, direction, can alias. Power Doppler: amplitude only, no direction, no aliasing, more sensitive to slow flow. Wall filter removes low-frequency clutter (vessel wall motion). Processing: spectral PW and CW use FFT; color uses autocorrelation. Color packet (ensemble) size up: more accurate velocities, lower frame rate; a wider color box also lowers the frame rate. Velocity error grows fast as the angle widens: keep the angle low and correct for it. Tissue Doppler: strong, slow echoes from the myocardium; wall filter minimal.

EXAMPLE Shift, Nyquist, aliasing

f_0 = 5,000 kHz, 0.5 m/s, cos θ = 0.5: $\Delta f \approx$ 1.62 kHz. PRF 5 kHz ⇒ Nyquist = 2.5 kHz: 1.62 kHz is below 2.5 kHz, so no aliasing. Highest velocity before aliasing (f_0 in Hz, c in m/s): v_{max} = 0.77 m/s.

Hemodynamics

$$Q = \frac{\Delta P}{R} \quad R = \frac{8\eta L}{\pi r^4} \quad Q = Av$$

Stenosis: velocity up, pressure down (Bernoulli); flow is conserved. Simplified Bernoulli: $\Delta P \approx 4v^2$ (mmHg, m/s). Resistive index: RI = (PSV – EDV)/ PSV. Plug flow: flat profile, narrow spectrum. Laminar (parabolic): center fastest. Turbulence: spectral broadening, fills the window. High-resistance bed: triphasic waveform; low-resistance bed: forward flow in diastole.

EXAMPLE Gradient and RI
v = 3 m/s: $\Delta P = 4 \times 3^2$ = 36 mmHg. PSV 100, EDV 30 cm/s: RI = 0.7.

Artifacts

Artifact	Looks like	Cause
Reverberation	equally spaced parallel lines deep to a strong reflector	sound bouncing between two strong reflectors (or reflector and transducer)
Comet tail / ring-down	short bright tapering trail behind a tiny strong reflector or gas	reverberations too close to resolve (comet tail) or resonating gas bubbles (ring-down)
Mirror image	duplicate placed deeper than a strong specular reflector, e.g. the diaphragm	the beam is redirected by the reflector; the system assumes a straight path
Acoustic shadowing	dark band deep to a strongly attenuating or reflecting structure	high attenuation: stone, bone, gas
Acoustic enhancement	bright band deep to a weakly attenuating structure	low attenuation: fluid-filled cyst, bladder
Edge shadow	thin shadow from the curved edge of a round structure	refraction and divergence at the curved surface
Refraction duplication	structure displaced sideways or duplicated	oblique incidence on a speed boundary (Snell)
Side lobe / grating lobe	off-axis echoes placed on the main beam axis	energy outside the main beam; grating lobes come from arrays
Speed error (range error)	reflector shown too deep or too shallow; step-off of a straight line	true speed differs from the assumed 1540 m/s
Slice thickness	false echoes filling anechoic structures	finite beam width in the elevational plane
Range ambiguity	deep echo shown too shallow	echo returns after the next pulse is sent (PRF too high for depth)

If X doubles

If	Then
frequency ×2	period ÷2, wavelength ÷2, SPL and PD ÷2 (same cycles), axial resolution number ÷2 (better), attenuation in dB ×2, penetration ÷2, near zone length ×2, Doppler shift ×2, aliasing velocity limit ÷2
imaging depth ×2	PRP ×2, PRF ÷2, duty factor ÷2, frame rate ÷2, Nyquist limit ÷2
cycles per pulse ×2	PD ×2, SPL ×2, axial resolution number ×2 (worse), duty factor ×2
element diameter ×2	near zone length ×4, divergence in the far field less
amplitude ×2	power ×4, intensity ×4
power ×2 (same area)	intensity ×2 (+3 dB)
beam area ÷2 (same power)	intensity ×2
propagation speed ×2 (same f)	wavelength ×2, PRP ÷2

EXAMPLE Speed error
True depth 5 cm in tissue at 1,450 m/s (system assumes 1,540): shown at 5 × 1,540 / 1,450 ≈ 5.3 cm (too deep: slower medium).

Bioeffects & safety

CLINICAL NOTE ALARA
Use the lowest output and shortest scan time that give a diagnostic image; monitor TI and MI on screen (AIUM).

- **Thermal**: absorption; TI ≈ estimated maximum temperature rise (°C) under modeled conditions. TIS soft tissue, TIB bone at the focus (fetal), TIC cranial bone at the surface.
- **Mechanical** (cavitation): stable (bubbles oscillate) or transient/inertial (bubbles collapse). Gas bodies (lung, bowel, contrast) are most at risk.

$$\text{MI} = \frac{p_r}{\sqrt{f}}$$

p_r derated peak rarefactional pressure (MPa), f in MHz. US FDA Track 3 limits (non-ophthalmic): derated $I_{SPTA} \le 720$ mW/cm², MI ≤ 1.9.

EXAMPLE MI
p_r = 2 MPa, f = 4 MHz: MI = $2/\sqrt{4}$ = 1.0.

Quality assurance

Tissue-mimicking (gray-scale) phantom: axial and lateral resolution, depth (registration) accuracy, dead zone, sensitivity and penetration, contrast, cyst imaging. Doppler: flow phantom or moving-string phantom (velocity accuracy, gate location). Hydrophone: measures output (pressure, intensity).

$$\text{Sens} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad \text{Spec} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

PPV = TP /(TP + FP); NPV = TN /(TN + FN).

EXAMPLE Test accuracy
TP 90, FN 10, TN 80, FP 20: sensitivity = 90%, specificity = 80%.

TRAP

- Propagation speed is set by the medium, never by frequency or the sonographer.
- Axial resolution: **smaller** number = **better**.
- Lateral resolution is best at the focus, not at the face of the probe.
- Deeper imaging: lower PRF, lower Nyquist limit, more aliasing.
- Gain does not change patient exposure; output power does.
- No refraction at normal incidence, even with different speeds.
- Doppler at 90°: no shift, not “zero flow”.
- Power Doppler shows no direction and does not alias.
- Pulse–echo attenuation counts the round trip (2 × depth).
- Mirror image sits **deeper** than the true structure.
- Shadowing: high attenuation; enhancement: low attenuation.
- dB are logarithmic: +3 dB doubles the intensity, it does not add a few percent.